Conditional Statements — Practice

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Hypothesis and Conclusion

Watch Me — Worked Example

"If two angles are vertical angles, then they are congruent."
Hypothesis (p): two angles are vertical angles. Conclusion (q): they are congruent.
Not in if-then form? "All squares are rectangles" $\rightarrow$ If a shape is a square, then it is a rectangle.

1. "If $x = 5$, then $2x = 10$." Name the hypothesis.

_____

2. "If a figure is a triangle, then it has three sides." Name the conclusion.

_____

3. Write in if-then form: "All right angles measure $90°$."

_____

4. Write in if-then form: "Fish live in water."

_____

Type 2 Converse, Inverse, Contrapositive

Watch Me — Worked Example

Conditional: If an animal is a dog, then it has four legs.
Converse (switch): If an animal has four legs, then it is a dog.
Inverse (negate): If an animal is not a dog, then it does not have four legs.
Contrapositive (switch and negate): If an animal does not have four legs, then it is not a dog.

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5. "If you live in Los Angeles, then you live in California." Write the converse.

6. "If an angle measures $120°$, then it is obtuse." Write the contrapositive.

7. "If $x = 3$, then $x^2 = 9$." Write the inverse.

8. p: it is Saturday. q: there is no school. Write $\sim q \rightarrow \sim p$ in words.

Type 3 True or False? Give a Counterexample

Watch Me — Worked Example

"If $x^2 = 9$, then $x = 3$." Try $x = -3$: $(-3)^2 = 9$, but $x \neq 3$.
Answer: false — counterexample $x = -3$. (To show a statement is false, one counterexample is enough.)

9. "If a number is divisible by 10, then it is divisible by 5." True or false?

10. Converse of #9: "If a number is divisible by 5, then it is divisible by 10." True or false? Counterexample?

11. "If two angles are supplementary, then they form a linear pair." True or false?

12. A conditional is true. Must its contrapositive be true? Must its converse be true?

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Type 4 Biconditionals

Watch Me — Worked Example

"If an angle is a right angle, then it measures $90°$" is true, and so is its converse.
Biconditional: An angle is a right angle **if and only if** it measures $90°$.

13. Write as a biconditional: "If two angles have the same measure, then they are congruent" (the converse is also true).

14. Is this a good biconditional? "An angle is acute if and only if it measures $40°$." Explain.

15. Write the two conditionals in: "$x + 2 = 7$ if and only if $x = 5$."

16. HARD A statement is false. Which related statement must also be false: converse, inverse or contrapositive?

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